

On Pastiche, Distributions, and Appropriations in Writing Assistants: A Squib

Jaylen Pittman
Stanford University

CCS CONCEPTS

• Computing methodologies → Natural language generation.

KEYWORDS

sociolinguistics, discourse, pragmatics

1 THE PASTICHE

In his provocatively titled blog post "How come GPT can seem so brilliant one minute and so breathtakingly dumb the next?", Gary Marcus [9] offers the metaphor of the *pastiche* as a means through which to understand the oscillations in apparent reasoning behavior in OpenAI's ChatGPT. A pastiche, being merely an imitation of form and not of not content, can seem to make sense to sense-making humans. Even slightly complex semantic, pragmatic, and discourse level tasks easily convince the human that the pastiche language model is not merely a stochastic parrot [1], but also a far cry from any reasonably reliable interlocutor.

Marcus [9] attributes model hallucinations in GPT 3.5 partially to a problem of *binding*, wherein entities and their (semantic, world-grounded) properties are somewhat dissociated in the course of training neural networks, and a related point is made in the cognitive science of language literature via appeal to a formal/functional competence distinction common to generative linguistics [8]. While a pastiche language model might foster greater creativity or perhaps inspire Dadaists [7], the implications for writing assistants which engage with minoritized language varieties are severe. Imitation of language features in human conversation often hinge on power asymmetries in interaction and tend to reproduce negatively racialized stereotypes of ethnic and/or racial groups [6, 12].

2 DISTRIBUTIONS AS LANGUAGE PROFILES

Within linguistics, the subfields of language typology and sociolinguistics are concerned with the distributional profiles of languages and their features [2, 5, 10, 11]. Despite different aims, research in these areas demonstrate that the mere presence or absence of a feature of a language variety (say, consonant coda reduction) provides only surface-level information relevant to that variety. Rather, one must look at the rates of use of that feature, and how those features are enabled or constrained by other features surrounding it. What tends to happen in large language models that attempt to write in language varieties such as African American English is what Fix calls hyperperformance, in which a narrow set of language features are used at rates disproportionate to actual usage by members of the speech community [3, 4].

3 IMPLICATIONS FOR WRITING ASSISTANTS

My fear is that writing assistants applied to minoritized language varieties such as African American English might reproduce negative

stereotypes through hyperperformance of particular features of AAE at the expense of others, while also not taking into account the distributional profile of the language's various resources. The inequity of the situation might be further amplified by creatives who are non-native speakers of the variety who are unfit to judge the well-formedness of the generated output, thereby perpetuating mock Ebonics [12].

REFERENCES

- [1] Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?. In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency*. 610–623.
- [2] Balthasar Bickel. 2015. *Distributional Typology: Statistical Inquiries into the Dynamics of Linguistic Diversity*. Oxford University Press, 0. <https://doi.org/10.1093/oxfordhb/9780199677078.013.0046>
- [3] Maeve Eberhardt and Kara Freeman. 2015. 'First things first, I'm the realest': Linguistic appropriation, white privilege, and the hip-hop persona of Iggy Azalea. *Journal of Sociolinguistics* 19, 3 (2015), 303–327. <https://doi.org/10.1111/josl.12128>
- [4] Sonya Fix. 2010. Representations of blackness by white women: Linguistic practice in the community versus the media. *University of Pennsylvania Working Papers in Linguistics* 16, 2 (2010), 8.
- [5] Matthew K. Gordon. 2016. *Phonological typology*. Vol. 1. Oxford University Press.
- [6] Jane H. Hill. 1998. Language, race, and white public space. *American anthropologist* 100, 3 (1998), 680–689.
- [7] Gunter Loesel, Piotr Mirowski, and Kory Wallace Mathewson. 2020. Do Digital Agents Do Dada?. In *ICCC*. 488–491.
- [8] Kyle Mahowald, Anna A. Ivanova, Idan A. Blank, Nancy Kanwisher, Joshua B. Tenenbaum, and Evelina Fedorenko. 2023. Dissociating language and thought in large language models: a cognitive perspective. *arXiv:2301.06627* (Jan 2023). <https://doi.org/10.48550/arXiv.2301.06627> arXiv:2301.06627 [cs].
- [9] Gary Marcus. 2022. How come GPT can seem so brilliant one minute and so breathtakingly dumb the next? <https://garymarcus.substack.com/p/how-come-gpt-can-seem-so-brilliant>
- [10] John R. Rickford. 1987. The haves and have nots: Sociolinguistic surveys and the assessment of speaker competence. *Language in society* 16, 2 (1987), 149–177.
- [11] John R. Rickford. 2004. Implicational scales. *The handbook of language variation and change* (2004), 142–167.
- [12] Maggie Ronkin and Helen E. Karn. 1999. Mock Ebonics: Linguistic racism in parodies of Ebonics on the Internet. *Journal of sociolinguistics* 3, 3 (1999), 360–380.